Thinking Sideways, Observed: Interpreting the Lateral-Thinking Search Behavior of LLM Agents

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Complete draft — no open gaps.

Abstract

Give a language-model agent thirty rounds of yes/no questions to crack a lateral-thinking puzzle and it does not keep getting better: on 22 human-played Turtle Soup puzzles, checkpoint accuracy improves only in the first ten rounds and only at scale (Qwen3.5-397B $0.12 \rightarrow 0.20$, then flat; a 4B model stays at 0.05 throughout), and extending the round budget monotonically *hurts* the small model ($0.050 \rightarrow 0.004$). Outcome scores cannot say why, so we read the process: each game's questions form a trajectory in association space whose geometry separates two failure modes that share a score of zero. The small model stalls — its late-game stride shrinks to under half the mid-size model's (0.030 vs 0.082) and it volunteers a final story in 2 of 66 games — while larger models stride twice as far, bank their gains early, and then stop converting exploration into accuracy. We are equally plain about what the geometry does not do: within puzzles, mixed-effects models find no predictive power for stride or drift (LRT $p \geq 0.20$), unchanged under a surface-only anchor—an interpretive instrument, not a score predictor. One pre-experiment lesson: an unaudited Oracle collapses questions to "irrelevant," making every game unwinnable, so agent claims begin with an information audit of the environment. We release the harness, the provenance-enforced puzzle set with an under-determination difficulty measure, and a two-part score whose halves demonstrably measure different competences.

1 Introduction

Here is a Turtle Soup puzzle. *A man's body lies in the desert, clutching half a matchstick; luggage and clothing are scattered around him.* That is all the solver sees. The hidden story is that a hot-air balloon was losing altitude, the passengers threw out their luggage and still could not stay up, so they drew lots with matchsticks—and the man who drew the short one was thrown out. The solver recovers this by asking yes/no questions, one per round.

Outcome metrics can say whether an agent got there—and on our verified set that story is flat: accuracy rises only in the first ten of thirty rounds, only for the larger models, and twenty further rounds buy nothing at any scale we tested (§5). What outcome metrics cannot say is how the agent searched. Two agents scoring 0.0 on the same puzzle may be failing in opposite ways: one never leaves its first hypothesis, the other leaves the plausible region entirely. Telling them apart requires instruments that read the process, and that is this paper's contribution.

Why Turtle Soup. The questioning protocol is a naturally verbalized search trace—no probing or activation access needed, the behavior *is* text; puzzles are deliberately under-determined, so the direction of each associative jump is diagnostic; and games are short, language-only, and cheap to run at scale.

Central hypothesis. An agent's per-round questions, embedded in a semantic space, form a trajectory whose geometry explains success and failure.

- **H1 (stalling).** An agent stuck in a hypothesis basin shows step size going to zero; lexical novelty heuristics catch only the verbatim extreme of this.
- **H2 (drifting).** An agent exploring where no human path goes shows normal step size but growing anchor distance—invisible to lexical novelty, because synonym-shuffled re-asks look new.
- **H3 (predictivity).** Trajectory features predict final solve quality beyond outcome-adjacent covariates.

Contributions. (1) The flatness result—interaction length buys accuracy only in the first $\sim$10 rounds and only at scale, and budget extension actively harms the smallest model. (2) A behavioral-interpretation framework separating stalling from unbanked exploration, with an honest account of what it does and does not predict. (3) An open-source harness, a **22-puzzle set restricted to puzzles humans have played and solved**, and a composite score decoupling clue recall from judged causal logic. (4) A puzzle-level difficulty measure, **under-determination**, independent of rounds used—so it can group an accuracy-versus-rounds analysis without circularity.

2 Related Work

Existing turtle-soup benchmarks—TurtleBench [1], SPLAT [2] (whose interactive player–judge protocol we adopt), and TurtleSoup-Bench [3]—score verification accuracy, solve efficiency, or faithfulness to the intended story: the destination, where we score the trajectory. MUTATE [4] shows frontier agents find far fewer mechanism-distinct paths than humans; its diverge-then-narrow agent is a natural intervention to test against our metrics. The instrument itself adapts forward flow [7] and the Divergent Association Task [8] from human psychometrics—semantic travel as a measure—with the Small World of Words norms [9] as the human anchor ours is not yet (§3). Judge-bias catalogues [5, 6] shape the scoring: the model judge is confined to the one subscore objective matching cannot reach.

3 Framework

3.1 Task and harness

A **Questioner** sees only the 汤面 and asks one yes/no question per round. An **Oracle** holds the 汤底 and answers 是/不是/与此无关. The Questioner may commit a final story at any round, or is forced to at the budget. The harness has pluggable providers—local, gateway, and token-billed sampling including RL-trained checkpoints—with full per-round logging and per-game seeds.

3.2 Puzzles

A puzzle nobody has solved carries no evidence that its clues suffice, that its solution is unique, or that the intended leap is reachable—an agent scoring zero on one teaches us nothing. Our set therefore takes only puzzles with a public record of human play: 22 items from a Chinese Turtle Soup community site, each keeping its source URL. Surfaces run 9–108 characters (median 34), solutions 13–200 (median 102), with 97 annotated key clues. Provenance is enforced in code—a verified family and a quarantined generated family, with tests that fail if the two mix.

Each puzzle carries an **under-determination** score: how far the surface alone leaves you from the solution, measured by sampling twelve cold guesses from the surface with no Oracle feedback and taking the closest. It ranges 0.173–0.577 (median 0.347) over the set and is *not* explained by surface length ($r = -0.22$, n.s.), so it is available as a puzzle-level covariate—the thing the mixed-effects model in §5 previously lacked.

3.3 Outcome scoring: composite, not surface

Judged outcome is **clue recall (70)** plus **causal-logic identity (30)**; Appendix B gives the detail. String matching alone cannot tell a wrong answer from a right one worded differently—we watched a frontier model reconstruct a hidden causal chain completely and score 0.00. Conversely the clue half blocks “far = creative” gaming: distance without validity scores zero.

3.4 Association-trajectory instrumentation

Each round, we extract the question’s keywords, embed them, and aggregate to a unit vector q_t. Two signals follow: **step size** $s_t = d(q_t, q_{t-1})$, the agent’s associative stride, and **anchor distance** $h_t = d(q_t, \mathcal{A})$. Hypothesized signatures: stalling is $\bar{s} \to 0$ with h flat; drifting is normal $\bar{s}$ with h_t rising. The lexical `question_novelty` metric is kept as the ablation baseline H2 predicts will miss synonym-drift.

On the anchor, stated plainly. Ours is built from the puzzle’s surface, solution and clues—**not** a human association manifold, and we no longer describe it as one. Its terms overlap the clue set that defines the outcome, and rebuilding it from the surface alone flips the drift slope’s sign on a minority of traces (quantified in §5); we therefore report both anchors, and treat a SWOW-based version [9] as the correct future form.

4 Experiment Designs

E1 — Round curve. Play to 30 rounds; after each round force a checkpoint answer and score it. Does interaction help, and where does behavior degrade?

E2 — Round budget. Hard caps $\{5, 10, 15, 20, 25, 30\}$ with a forced final answer. How does behavior change under pressure?

E3 — Association trajectory. For every E1 game compute $\{s_t\}$ and $\{h_t\}$ and test H1–H3. The keyword extractor is held fixed across models, so no model is scored through its own vocabulary.

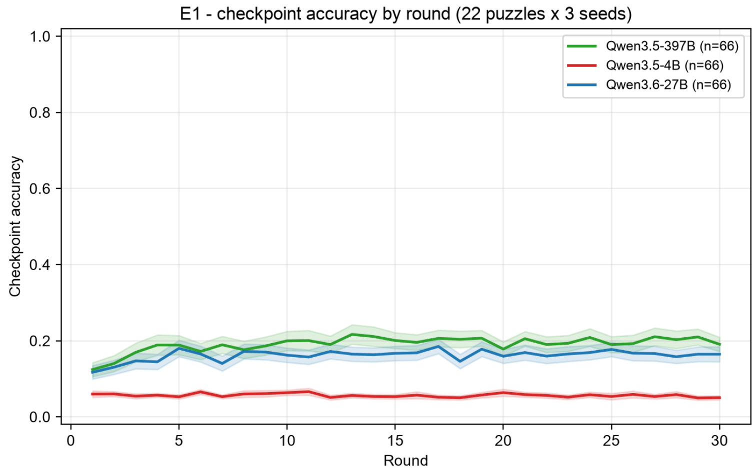

Figure 1: E1: mean checkpoint accuracy by round (22 puzzles $\times$ 3 seeds per model), on the full 0–1 outcome range so that flatness reads as flat. Gains exist only before round $\sim$10 and only at scale; the 4B curve never moves.

E4 (exploratory). Diverge-then-narrow prompting [4]—which trajectory feature does it move? And do signatures shift on puzzles admitting many distinct stories versus one?

Prerequisite — the Oracle audit. The Oracle is the sole information channel: a weak one collapses answers to "irrelevant"—the reply carrying no information—after which no Questioner can succeed and every score measures the environment. Candidates are probed on held-out pairs before any agent claim, and **must come from a different model family than the Questioners**, or a same-family Questioner's advantage cannot be separated from its scale.

5 Results

All numbers come from one grid on the verified set: 22 puzzles $\times$ {Qwen3.5-4B, Qwen3.6-27B, Qwen3.5-397B-A17B} $\times$ 3 seeds, with a DeepSeek-V3.1 Oracle and logic judge (cross-family, audited at 95%), and zero failed games. E1 is 198 thirty-round games with a scored checkpoint every round; E2 is 1,188 games across caps $\{5, 10, 15, 20, 25, 30\}$.

5.1 More interaction stops helping after ten rounds (E1)

Checkpoint accuracy rises only in the first $\sim$10 rounds, and only for the larger models: Qwen3.5-397B climbs from 0.124 (round 1) to 0.200 (round 10) and then stays between 0.18 and 0.21 through round 30; Qwen3.6-27B rises $0.117 \rightarrow 0.162$ and plateaus near 0.16. Qwen3.5-4B is flat for the entire game, 0.05–0.06 at every checkpoint (Figure 1, drawn on the full 0–1 range). Twenty further rounds of yes/no evidence—two thirds of the interaction budget—buy no accuracy at any scale.

Commitment separates the models more than accuracy does. 27B volunteers a final story in 11/66 games (mean round 11.1) and 397B in 9/66 (round 14.2); 4B commits in 2/66 (round 23.5)—it neither improves nor concludes.

5.2 Budget pressure is asymmetric (E2)

End accuracy is insensitive to the round cap for the larger models (397B 0.19–0.21 and 27B 0.14–0.18 across all six caps), but **monotonically harmful for the smallest**: 4B falls from 0.050 at cap 5 to 0.004 at cap 30, with 175 of its 396 games ending by token-budget exhaustion rather than by answer (versus 2/396 for 27B and 0/396 for 397B). Longer games do not help a small model; they give it more rounds in which to fail.

The two score halves dissociate. Every model earns proportionally far more of the judged causal-logic half than the clue-recall half: 397B averages 12.6/30 logic against 7.2/70 clues, 27B 10.1 against 6.8, 4B 2.9 against 0.2. Models reach the right causal shape well before—and often without—landing the annotated clue content, so the two subscores measure different competences and the composite's logic half is not redundant with string-level recall.

5.3 Geometry explains the plateau but does not predict the score (E3, H3)

The models occupy different geometric regimes: mean associative stride is 0.049 for 4B against 0.088 (397B) and 0.114 (27B), and in the late game 4B's stride falls to 0.030—under half of 27B's 0.082—while it almost never commits. That

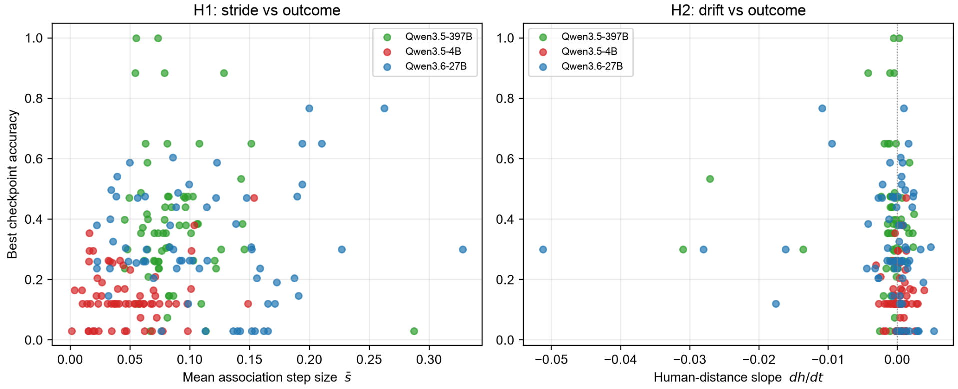

Figure 2: E3: per-trace geometry against best outcome ($n = 198$). Left: mean stride $\bar{s}$; the models separate into regimes (4B small strides and low scores), but within a model stride does not predict outcome. Right: drift slope dh/dt, clustered near zero with no within-model relationship—the visual form of H3's negative result.

is H1's stalling signature: the small model circles a hypothesis basin with shrinking steps. Pooled across models, stride correlates with best checkpoint score (Spearman $\rho = 0.23$, $n = 198$); but this is a between-model effect—**within** each model the correlation vanishes ($|\rho| \leq 0.16$ for every model and every geometric feature; Figure 2).

The mixed-effects test makes that precise. With puzzle as a random intercept and model tier as a fixed effect, adding the drift slope does not improve fit (LRT $p = 0.39$), nor does stride $\times$ tier ($p = 0.20$); the puzzle intercept absorbs the variance the geometry seemed to carry. The new puzzle-level covariate sharpens the interpretation: under-determination points the expected way (harder surface $\rightarrow$ lower score, coefficient -0.031) but is not significant ($p = 0.22$) and removes only $\sim$8% of the puzzle variance—the puzzle effect is not merely measured difficulty. **H3 is not supported within-puzzle**, and we accordingly claim trajectory geometry as an interpretive instrument—it says *how* an agent is failing—not as a score predictor.

Both anchors, as promised (§3). Drift slopes computed from the solution-aware and the surface-only anchor agree in rank ($\rho = 0.80$) yet flip sign on 38/198 traces; the mixed-effects conclusion is the same under both (surface-only drift LRT $p = 0.10$). The anchor choice changes individual traces, not the finding.

6 Harness Observations

Two observations, each a trap for benchmarks of this shape rather than a claim of ours.

Stalling is real, and lexical metrics only half-see it. A small model re-asks earlier questions *verbatim* in later rounds. A lexical novelty metric catches that—but only because it is verbatim. Earlier the same model re-asks one hypothesis in fresh words, which that metric scores as new and the geometry does not.

Truncated thinking masquerades as behavior. Under too small a token budget, a reasoning model's logged "questions" are fragments of chain-of-thought, and every Oracle reply to them is "irrelevant." From the transcript this is indistinguishable from an agent that cannot form a question. Decoding configuration is part of the environment.

7 Limitations and Conclusion

Limitations. The anchor is not yet human-derived (§3), so H2's framing is descriptive until SWOW norms or collected human traces are in place; we do not equate human-unlike with wrong. Keyword extraction and encoders are themselves models; feedback incorporation—the coordinate that would separate stalling from unbanked exploration most directly—is definable but unmeasured. The source site is public, so memorisation probes are required rather than optional: the two puzzles our difficulty measure rates easiest are the two most widely circulated. Puzzles involve death and dark themes; content-flagged.

Conclusion. Turtle Soup turns hypothesis-space search into observable text, and trajectory geometry turns that text into behavioral signatures. An endpoint score projects the whole search onto one number, destroying exactly what improving an interactive agent requires: a stalling agent and one that explores without banking need opposite interventions, and reading the trajectory is how you tell which one you have.

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A The puzzle set

Where the puzzles come from. A puzzle nobody has solved carries no evidence that its clues suffice, that its solution is unique, or that the intended leap is reachable — an agent scoring zero on one teaches us nothing. Our set is hand-picked from the classic repertoire of a Chinese Turtle Soup community: 22 puzzles people have played and solved, each keeping its source record. Every item was read and kept or rejected by hand; rejected were puzzles turning on deduction from stated evidence rather than a lateral reframing, and puzzles whose surfaces were incomplete at the source. Surfaces run 9–108 characters (median 34), solutions 13–200 (median 102).

How hard are they? Each puzzle carries an **under-determination** score: give a model the surface alone, with no Oracle and no feedback, let it write twelve complete stories, and take the closest to the real solution. The index is one minus that best guess — large when the surface leaves you far from the answer.

Under-determination	Puzzles
0.15–0.25	5
0.25–0.35	8
0.35–0.45	4
0.45–0.55	4
0.55–0.65	1

Median 0.347, quartiles 0.273 and 0.410, full range 0.173–0.577 — single-peaked with a thin hard tail. Easiest are the 海龟汤 story itself (0.17) and the desert matchstick (0.19). Hardest is a fourteen-character surface reading *our heights differ / a flowerpot broke / our heights are the same* (0.58).

Two properties make this usable as a covariate. It describes the puzzle, not any agent's performance on it. And it is independent of rounds used — a difficulty defined as "how many rounds this takes" would make an accuracy-versus-rounds plot confirm itself. It is also not a proxy for surface length ($r = -0.22$, n.s.), so it carries information the obvious surrogate does not. One measure rather than two: a surface loose enough to admit many mechanisms is, for that same reason, one whose true mechanism takes more reframing to reach, so *how far* and *how many ways* are not separable here.

A caveat we report either way. The two puzzles scoring easiest are the two most widely circulated ones. Their surfaces may be easy to complete not because the inference is short but because the story is in the training data, in which case the measure reads familiarity as much as difficulty. A memorisation probe — asking for the solution with no surface given — separates the two and belongs in any use of this measure.

B Measuring agent performance

What counts as a correct answer. Judged outcome is **clue recall (70)** plus **causal-logic identity (30)**. Clue recall matches the answer against per-puzzle annotated key clues: objective and reproducible. The logic judge rates only whether cause → mechanism → outcome matches, is told explicitly to ignore wording, and is sampled three times and averaged, since single ratings are unstable on borderline answers. The gap between the two subscores is itself readable: right story, wrong vocabulary.

Clue matching tolerates paraphrase through character-bigram recall, which sets a floor on how short an annotated clue may be — below it only a verbatim match counts, and the score reads vocabulary rather than content. Clues are annotated above that floor.

Does the logic half earn its thirty points? On the full grid, yes: the two halves dissociate for every model (§5), with each model earning a several-fold larger share of the logic half than of the clue half. If the two rankings had agreed, clue recall alone would have been the cleaner scale; they do not.

Where the accuracy curve comes from. E1 does not score once at the end. Every round, after the Oracle answers, the Questioner is required to commit its best complete story, and *that* is scored — so each point on the accuracy curve is a full answer written with the evidence available at that round.

One game, worked. Puzzle `refsoup_021` (surface: a stormy night, a run-down temple, two photographs on the wall that seem to stare; by morning the narrator sees only two windows; solution: 墙上根本没有照片，是有人贴着窗户往里看), played by Qwen3.5-397B, seed 2. Table 1 shows six of thirty rounds; every cell is verbatim from the log, stories abridged to the clause carrying the hypothesis.

Three things the table shows that no mean curve can. **The checkpoint mechanism surfaces knowledge the agent never acts on:** the round-10 committed story *is* the solution—synthesized from earlier answers, on a round whose

轮	Question (abridged)	答	Committed story (abridged)	score (clue+logic)
1	照片里的是人吗?	是	那两张“照片”……是两个被钉死或封在墙洞里的人	0.12 (0+12)
3	主角第二天醒来时，还在那座寺庙里吗?	是	所谓的“两张照片”其实是两扇窗户……昏暗光线下……误看成了照片	0.91 (70+21)
10	“两扇窗户”是因为相框玻璃破碎脱落、变成通透的洞口吗?	不是	那根本不是照片，而是……躲在窗户后面（或贴在窗玻璃外侧）的人脸	**1.00** (70+30)
11	是因为照片被取走、露出了后面原本就存在的窗户吗?	不是	昨晚“盯着他的人像”，其实是窗外的人	**1.00** (70+30)
19	照片里的人像实际上是躲在照片后面的真人吗?	不是	窗户洞口被画着人像的布帘封住……雷雨导致遮挡物脱落	0.12 (0+12)
30	是因为积水使主角身体位置升高吗?	不是	两个被纸糊住的窗户洞口……雷雨导致纸张脱落	0.12 (0+12)

Table 1: Six rounds of one E1 game (`refsoup_021`, Qwen3.5-397B, seed 2): the Questioner's question, the Oracle's answer, the story the agent was forced to commit that round, and its composite score.

own question was answered 不是—yet the model never volunteers a final answer in all thirty rounds (one of the 57/66 non-committing 397B games), and by round 19 it has abandoned the correct reading for a falling-coverings mechanism it never escapes. **Stalling in fresh words happens at the largest scale too:** rounds 22–30 are eight lexically distinct re-asks of the same flooding hypothesis, exactly the pattern lexical novelty misses. **And the curve's round-to-round jitter is a scoring fact, not a knowledge fact:** the same understanding re-worded (rounds 3 $\rightarrow$ 4) swings the clue half 70 $\rightarrow$ 0; the mean curves of §5 average this out, which is why single-game curves are not the unit of analysis.